

Pre-Thesis I Report

Multi-Task Deep Learning Framework for Unified Cattle Health and Behavior Monitoring

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A thesis submitted to the Department of Computer Science and Engineering
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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.
2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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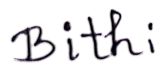
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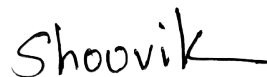
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Approval

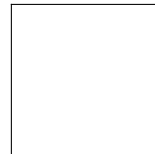
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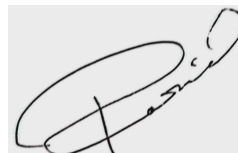
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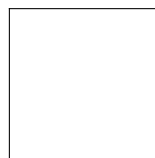
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Abstract

The fast development of precision livestock farming has created a need for systems capable of doing multiple health assessments. Current cattle monitoring research has a major problem: most studies only tackle one task at a time and use private data that other researchers can't access. This makes it nearly impossible to verify results or deploy these systems on real farms. This thesis takes a different approach. We developed a single deep learning system that handles four cattle monitoring tasks together: Body Condition Scoring, behavior recognition, lameness detection, and identifying individual animals. What sets this work apart is our commitment to transparency—we only use publicly available datasets, so anyone can reproduce and build on our work. Our approach uses a lightweight neural network (EfficientNet-B0 or MobileNetV3-Small) as a shared foundation. We train each task sequentially, which lets us combine different types of data without needing labels for every task in every image. To boost performance, we incorporated advanced techniques like CORAL (which helps with ordinal rankings) and CBAM (which helps the model focus on important features). While the system prioritizes RGB modalities, it also uses the contributions of depth information when available. This work provides the first complete, reproducible system that can run on edge devices to monitor four cattle health tasks in real-time across different farm environments.

Keywords: Multi-Task Learning, Cattle Health Monitoring, Body Condition Scoring, Behavior Recognition, Lameness Detection, Deep Learning, Transfer Learning, Public Datasets, Edge Deployment

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Nomenclature

Abbreviations

AI	Artificial Intelligence
AUC	Area Under the Curve
BCS	Body Condition Score/Scoring
BCE	Binary Cross-Entropy
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CBVD-5	Cow Behavior Video Dataset - 5 classes
CE	Cross-Entropy
CLAHE	Contrast Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CORAL	Rank Consistent Ordinal Regression
CPU	Central Processing Unit
DL	Deep Learning
DWT	Discrete Wavelet Transform
F1	F1-Score (harmonic mean of precision and recall)
FLOPs	Floating Point Operations
FPS	Frames Per Second
GAN	Generative Adversarial Network
GPU	Graphics Processing Unit
Grad-CAM	Gradient-weighted Class Activation Mapping
I3D	Inflated 3D ConvNet
ImageNet	Large-scale visual recognition dataset
IMU	Inertial Measurement Unit
IoT	Internet of Things
IoU	Intersection over Union
IR	Infrared
KD	Knowledge Distillation
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
mAP	mean Average Precision
Mask R-CNN	Mask Region-based Convolutional Neural Network
mIoU	mean Intersection over Union
ML	Machine Learning
MSE	Mean Squared Error

MTL	Multi-Task Learning
NMS	Non-Maximum Suppression
P1	Pre-Thesis I
PCA	Principal Component Analysis
PLF	Precision Livestock Farming
R-CNN	Region-based Convolutional Neural Network
ReLU	Rectified Linear Unit
ResNet	Residual Network
RF	Random Forest
RFID	Radio-Frequency Identification
RGB	Red Green Blue (color image)
RGB-D	RGB plus Depth
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROI	Region of Interest
RQ	Research Question
SE	Squeeze-and-Excitation
SORT	Simple Online and Realtime Tracking
SOTA	State-of-the-Art
SSL	Self-Supervised Learning
SVM	Support Vector Machine
t-SNE	t-Distributed Stochastic Neighbor Embedding
VGG	Visual Geometry Group
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once

Chapter 1

Introduction

1.1 Background

In terms of food security the livestock industry plays a vital role globally, where cattle is one of the most crucial livestock species economically worldwide. According to the Food and Agriculture Organization (FAO), the global cattle population exceeds 1 billion heads, contributing substantially to meat and dairy production **fao2023livestock**. The sustainability, income, and also productivity of farming tasks are all directly affected by the health and welfare of cattle. Early detection of health issues like poor body condition, lameness, and abnormal behavior can prevent economic losses estimated at billions of dollars annually while improving animal welfare standards **weary2009understanding**. Cattle health and nutritional status are fundamentally shown by Body Condition Scoring (BCS) . BCS is traditionally assessed through visual and tactile evaluation of fat deposits over key anatomical regions, typically using a 5-point scale with 0.25 increments **edmonson1989bcs**. The production of milk, the reproduction ability as well as capability to fight against the diseases are directly related with optimal body condition. Still, frequent monitoring is not an optimum solution for large scale operations because manual BCS assessment is very much time-consuming, subjective, and requires trained personnel Lameness represents another critical health concern, affecting up to 25% of dairy cattle in intensive farming systems **whay2003lameness**. Moreover, due to causing severe pain, lameness lowers the production of milk, the feed intakes decrease and also hinders the reproduction. But timely intervention is possible by early detection of abnormal body movements. Behavior monitoring highlights valuable insights into the health of cattle, stress levels, and welfare. Changes in feeding patterns, rumination time, lying behavior, and social interactions often precede clinical manifestations of disease **weary2009understanding**. Continuous 24/7 monitoring can be possible through automated behavior recognition that would be impossible through manual observation. Individual identification is fundamental to precision livestock farming, enabling tracking of health records, feeding behavior, and productivity metrics for each animal. The most common practices are using ear tags and RFID systems, but they require physical contact and can be lost or damaged. A non-invasive alternative that is offered by vision based identification can be integrated with health monitoring

systems. The convergence of computer vision, deep learning, and Internet of Things (IoT) technologies has created unprecedented opportunities for automated cattle health monitoring **neethirajan2020digitalization**. Convolutional Neural Networks (CNNs) have showed extraordinary success in image classification also object detection tasks, achieving human-level performance in different visual recognition challenges. Transfer learning opens the application of models pre-trained on large-scale datasets such as ImageNet to domains with limited labeled data. Inspite of these technological advances, already existing cattle monitoring systems predominantly achieve single tasks in isolation. A systematic review of 30+ research papers conducted for this thesis reveals that while multi-task learning is frequently mentioned, end-to-end implementations combining multiple health monitoring tasks are virtually non-existent. This fragmentation limits the practical utility of automated monitoring systems and fails to exploit the synergies between related tasks.

1.2 Rationale of the Study and Motivation

The motivation for this research stems from the major gaps in our current state of cattle health monitoring systems and technology, and the increasing need for precision livestock farming. 89% of the papers we have studied shows that the current research relies highly on private datasets, creating a massive reproducibility issue. Most of the current studies focus on individual problem solving like BCS, lameness, behaviour or identification- separately, without considering the benefits of Multitask Learning (MTL). We can significantly reduce computational cost while improving performance by sharing data across related tasks. This makes sense because the tasks we are focusing on are connected: abnormal body pose or gait pattern contribute to reduced feeding and declining body condition. Many published solution need expensive gear like thermal cameras, wearable sensors, depth sensors etc or controlled environments, which are not practical in real-world scenarios. This creates a clear need for RGB based solutions that can be operated with typical cameras while also using depth information if available. Current modern-farms need real-time processing on-site, where internet can be unreliable and cloud processing causes delay; the literature recognises edge deployment as a profound solution, but most solution rely on computationally demanding architectures that cannot be run on edge devices. We found that hardly anyone is using self-supervised learning for cattle monitoring, which is surprising, most farms already run surveillance cameras that record hours of footage daily. This unlabeled video just sits there, but it could actually help train better models—particularly useful when you don’t have much labeled data to work with. These problems—reproducibility, fragmented research, deployment limitations, and missed opportunities—show why we need a single, lightweight framework that tackles multiple cattle health monitoring tasks.

1.3 Problem Statement

Multiple fundamental problems remain unresolved in spite of significant research efforts in automated cattle health monitoring. Because of that, the practical impact of computer vision and deep learning solutions on actual farms is limited.

The major problem we found is that most systems treat tasks separately, thus requiring multiple systems to monitor multiple health assessments tasks. This leads to more computational load from running multiple models, not taking advantage of shared features that could help multiple tasks, complicated deployment because you need to integrate separate systems, and missing connections between tasks like how poor body condition often links to less feeding. Our systematic review confirms that there is no single system that combines BCS, lameness, behaviour and identification tasks in one framework. MLT is frequently mentioned but rarely implemented.

In order to check if results are correct or compare different methods properly, and improve on old research, the researchers need to use the same datasets—but most of the cattle monitoring studies use their own data which are private. This makes it very hard for others to reproduce the results, or to do fair comparison, or build on previous works in the proper way. Other computer vision domains don't face this issue since they have standard public benchmarks like ImageNet and COCO that everyone can use.

There are several papers that work on single-farm data and single-breed cattle. This raises questions regarding the generalizability of our proposed approaches. Due to specific farm environments, camera placement, or lighting models may overfit, and breed-specific appearance differences might not be captured. Seasonal and temporal changes are also hardly handled in evaluations. Cross-dataset evaluation, where models are trained on one dataset and tested on another, almost never happens, even though it is necessary for illustrating practical usefulness.

Although there is a lot of interest in depth imaging and thermal imaging for cattle monitoring in the literature, multimodal research is severely limited by the availability of public data. There are no public cattle thermal datasets worldwide, only MmCows offers synchronized IMU sensor data, and only two public datasets—MmCows and Dryad—offer depth modality. Due to this scarcity, researchers are forced to either gather confidential data or completely give up on multimodal approaches.

1.4 Objectives

The main purpose of this research is to use computer vision techniques to create a multitask, lightweight, edge deployable, and reproducible deep learning framework for cattle health and behaviour monitoring. The goal is creating a system that can handle multiple health related tasks at the same time. The important thing is that it only uses publicly available datasets, making the work reproducible by anybody.

To achieve this goal, the research sets out the following specific objectives:

1. **To develop a unified multi-task framework** capable of doing Body Condition Scoring, behavior recognition, lameness detection, and individual identification at the same time using shared backbone architecture
2. **To ensure full reproducibility** by conducting experiments on public datasets, which helps others to independently verify findings and perform their own research.
3. **To evaluate sequential task training strategies** that can handle Body Condition Scoring, behavior recognition, lameness detection, and individual identification all at once using a shared backbone with different heads for each task.
4. **To verify cross-dataset performance** by checking model’s performance across datasets from multiple farms, and conditions so that it can be proved that the model can handle real deployment environments.

These objectives address the reproducibility gap in agricultural AI research and create a scalable foundation for real-time cattle health monitoring across various farm environments.

1.5 Methodology in Brief

The proposed methodology follows a step-by-step process to build and test a multi-task deep learning framework for monitoring cattle health. The methodology has five main phases: data preparation, architecture design, sequential task training, evaluation, and ablation studies.

1.5.1 Data Preparation

The framework relies only on public datasets, grouped by task. For Body Condition Scoring (BCS), the datasets are the ScienceDB BCS Dataset, which has about 3000 RGB images labeled on a 5-point scale, the Dryad RGB+Depth Dataset with paired RGB and depth images for BCS, and the IPLAB BCS Database, which adds more RGB images mainly for validation. The MmCows Dataset provides RGB video with 7 behavior classes that includes standing, walking, feeding up and down, drinking, licking, and lying for behavior recognition, and the CBVD-5 Dataset provides RGB video with 5 behavior categories. The CattleLameness Dataset contains RGB video with lameness severity annotations, and the Mendeley Lameness Dataset adds more RGB data for cross-dataset evaluation for lameness detection. OpenCows2020 includes RGB images labeled by individual identity, Cows2021 provides RGB images from the Bristol dataset, and MmCows ID annotations contain individual labels for 16 Holstein cows for individual identification.

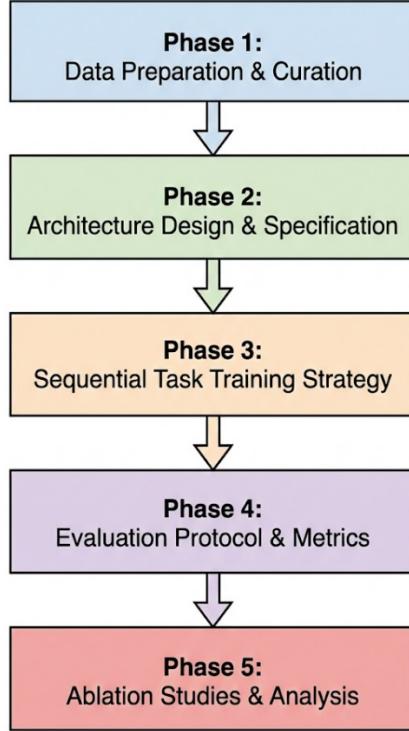


Figure 1.1: Methodology Workflow.

Data preprocessing includes resizing images to 224×224 (or 256×256 for depth), normalization using ImageNet statistics, data augmentation through random crop, horizontal flip, rotation ($\pm 15^\circ$), and CLAHE for lighting robustness, along with cow-wise splitting to prevent data leakage by ensuring individuals in the test set never appear in training.

1.5.2 Architecture Design

The multi-task model architecture has three main parts. First, the shared backbone uses either EfficientNet-B0 (5.3M parameters) or MobileNetV3-Small (2.5M parameters), and both are pre-trained on ImageNet, to learn general visual features that can support all tasks. Then, an attention module CBAM (Convolutional Block Attention Module) is inserted after the backbone to strengthen task-relevant features using both channel attention and spatial attention. The task-specific heads include a BCS Head which uses ordinal regression with CORAL loss so the class ordering is preserved, a Behavior Head which uses classification with focal loss to deal with class imbalance, a Lameness Head which performs either binary classification or multi-class classification, and an ID Head which uses metric learning with triplet loss or softmax classification.

1.5.3 Sequential Task Training

Training follows a sequential strategy that is supported by the MTL literature. In Phase 1, the backbone is initialized with ImageNet weights. In Phase 2, the backbone is frozen and the BCS head is trained using the ScienceDB and Dryad datasets. Phase 3 keeps the backbone frozen and trains the behavior head on MmCows and CBVD-5. Phase 4 continues with lameness head training on the CattleLameness dataset. Phase 5 trains the ID head on OpenCows2020. Then there is an optional Phase 6, where the backbone is unfrozen for joint fine-tuning using a weighted multi-task loss.

The multi-task loss function is defined as:

$$L_{total} = w_1 \cdot L_{BCS} + w_2 \cdot L_{behavior} + w_3 \cdot L_{lameness} + w_4 \cdot L_{ID} \quad (1.1)$$

where weights (w_1 to w_4) are tuned based on task importance and data availability.

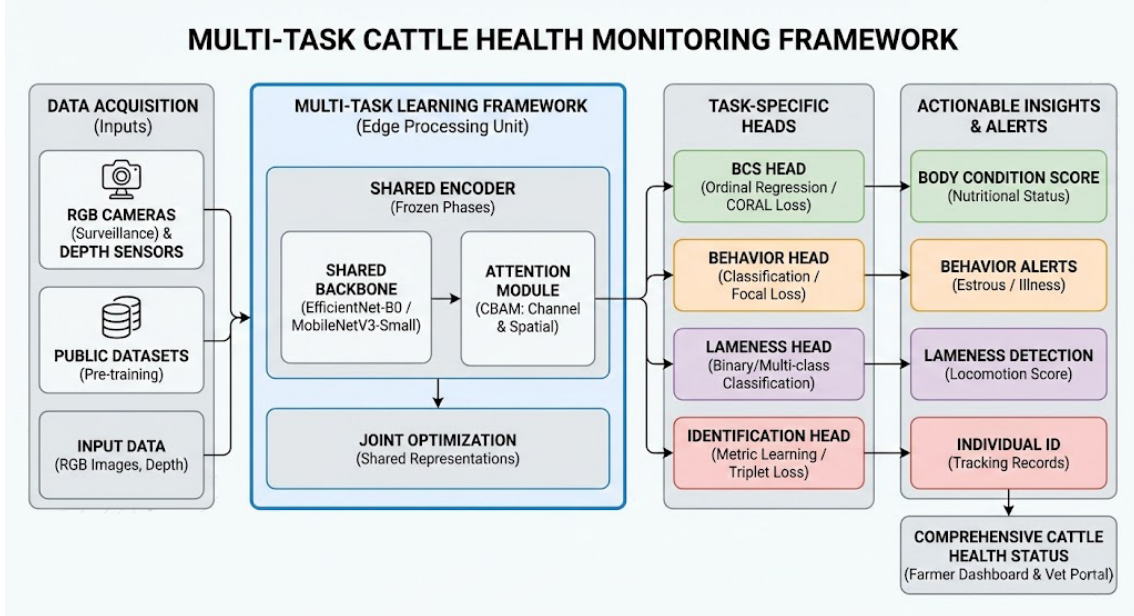


Figure 1.2: Conceptual Overview of Multi-Task Cattle Health Monitoring Framework.

1.5.4 Evaluation Protocol

Evaluation is matched to each task, since the outputs and risks are different. For BCS, the framework measures Mean Absolute Error (MAE) and accuracy within ± 0.25 and ± 0.5 tolerance. For Behavior, evaluation includes per-class accuracy, macro F1-score, and confusion matrix analysis. For Lameness, the metrics include sensitivity, specificity, and AUC-ROC. For ID, evaluation uses Top-1 and Top-5 accuracy along with mean Average Precision (mAP). Cross-dataset evaluation is also included by training on the main dataset and testing on a different dataset from another source, which helps measure generalization.

1.5.5 Ablation Studies

A set of structured ablation experiments will test the most important parts of the framework. For backbone selection, EfficientNet-B0 will be compared against MobileNetV3-Small and also ResNet-50 to identify which architecture works best. For loss functions, CORAL will be compared with Cross-Entropy for the BCS task. For training strategy, experiments will compare sequential versus joint versus progressive unfreezing. Through RGB-only versus RGB+Depth comparisons for BCS (using the Dryad dataset) the depth modality contribution will be tested. Finally, the attention mechanism will be tested by comparing results with and without CBAM.

1.6 Scope and Challenges

1.6.1 Scope of the Research

To ensure feasibility and depth of investigation this research stays within clear boundaries.

1. **Four Core Tasks:** Body Condition Scoring, behavior recognition, lameness detection, and individual identification, chosen mainly because public data exists for them and they also matter in practice.
2. **Public Datasets Only:** All experiments use publicly available datasets, and no private data collection is included, which supports full reproducibility.
3. **RGB as Primary Modality:** RGB images are the main input modality and depth is included only as a secondary modality for BCS (through the Dryad dataset).
4. **Dairy Cattle Focus:** The primary focus is on Holstein dairy cattle, since they dominate the available public datasets.
5. **Lightweight Architectures:** To ensure that the system stays suitable for edge development backbone selection is limited to models with $< 10M$ parameters.
6. **Sequential Task Training:** Frozen-backbone sequential training is the main training approach, with joint fine-tuning treated as an optional extension.

1.6.2 Challenges

Dataset Heterogeneity

Public cattle datasets are highly heterogeneous, including differences in resolution and image quality, camera angles including top-view, rear-view, and side-view, annotation protocols and how detailed the labels are, plus variation in breeds and farm environments. To reduce these issues, the strategies include standardized pre-processing pipeline, strong data augmentation, and domain adaptation techniques during joint fine-tuning.

Class Imbalance

Datasets of cattle monitoring suffer from high class imbalance, which is a significant issue. For lameness detection, lame individuals are outnumbered heavily by healthy cattle, in comparison to normal BCS scores (2.5-3.5), extreme BCS scores like (1-2 and 4-5) are much rarer, and some behaviors like drinking and licking happen infrequently. To address these issues, the architecture uses focal loss for classification tasks, weighted sampling during training, and class-balanced evaluation metrics.

Temporal Modeling for Video Tasks

From temporal context behavior recognition and lameness detection can benefit, but video-based modeling increases computational cost, and temporal models like LSTM and I3D add extra complexity and there can be inconsistencies between frame-level and clip-level annotation. The strategy involves initially using frame-level classification and apply temporal aggregation, and then treat LSTM integration as an optional extension depending on initial results.

Preventing Negative Transfer

Multi-task learning also risks negative transfer, where learning for one task hampers performance on another. This can happen when task gradients conflict during joint training, the best feature representations may vary depending on the task, and imbalanced task difficulties can dominate learning. The main mitigation is sequential training with a frozen backbone to separate task learning, along with gradient normalization and uncertainty weighting during joint fine-tuning.

Limited Labeled Data

Even though multiple public datasets are used, labeled data is still small compared to general computer vision benchmarks. ScienceDB BCS has around 3000 images, compared to millions in ImageNet. MmCows covers only 16 individuals over 2 weeks, and CattleLameness contains relatively limited lameness examples. To address this, the methodology depends on transfer learning from ImageNet, strong regularization (including dropout and weight decay), and data augmentation. Self-supervised pre-training on unlabeled frames is also included as a possible extension if it becomes necessary.

Chapter 2

Literature Review

2.1 Preliminaries

This chapter presents a systematic review of the research papers related to cattle health monitoring using computer vision and deep learning. The review covers five primary research areas: Body Condition Scoring (BCS), lameness detection, behavior recognition, individual identification, and cleanliness assessment. The study identifies current state-of-the-art methods, public datasets which are available, the research gaps, and suitable ways for developing a multi-task learning framework.

2.1.1 Multi-Task Learning Fundamentals

Multi-Task Learning (MTL) is a machine learning standard where a model is trained to multiple related tasks simultaneously, leveraging shared representations to improve generalization across all tasks **crawshaw2020mtl**. In deep learning terms, MTL generally uses a shared backbone network that takes out common features, with task-specific heads branching from the shared representation. The benefits of MTL include improved generalization through shared representations that act as regularization to reduce overfitting, computational efficiency via a single forward pass serving multiple tasks, data efficiency where tasks with limited data benefit from related tasks, and enhanced feature learning where auxiliary tasks can improve primary task features. Still, MTL also comes with some challenges like negative transfer when tasks interfere with one another, difficulties while balancing gradients also identifying the most efficient parameter sharing strategies. The Crawshaw 2020 MTL Survey provides a comprehensive taxonomy distinguishing between hard parameter sharing (common backbone) and soft parameter sharing (separate networks with regularization) approaches.

2.1.2 Body Condition Scoring Overview

Body Condition Scoring(BSC) is a standard way for evaluating the fat reserves of cattle through visual and tactile evaluation. The most common scale ranges start from 1 (skeletal) to 5 (fat) with 0.25 increments, giving up 17 possible scores. Key anatomical regions assessed include the tailhead and pin bones, hook bones and loin area, ribs and spine prominence, and hip bone region. Automated BCS using computer vision typically employs rear-view or top-view images capturing the tailhead region. The task can be worked out as classification (individual BCS classes), ordinal regression (preserving class ordering), or regression (continuous scores).

2.1.3 Lameness Detection Fundamentals

Lameness in cattle shows by identifying walking behaviour including reduced step length and walking speed, head movement and back curving, unwillingness to bear weight on the affected leg, and uneven hoof placement. Computer vision approaches for lameness detection typically analyze video sequences captured from side-view or top-view cameras. Key features include pose estimation of limb joints, gait cycle analysis, and motion trajectory patterns. Scoring systems range from binary (lame/sound) to multi-level severity scales (1-5).

2.1.4 Behavior Recognition Categories

Cattle behavior monitoring address numerous activities related to health and welfare, which includes feeding behaviors for example feeding, ruminating, and drinking; body movement activities such as walking, standing, and lying; social behaviors including licking, mounting, and aggression; also health indicators such as estrus behavior and calving signs. Vision-based behavior recognition implements detection (locating cattle in frames), tracking (maintaining identity across frames), and action classification (categorizing behavior). Temporal modeling by using LSTM, 3D CNNs, or attention mechanisms records behavioral changes.

2.2 Review of Existing Research

This section shows a comprehensive analysis of related research arranged by task category, followed by key multi-task learning and survey papers.

2.2.1 Body Condition Scoring Research

Body condition scoring represents the most highly studied task in cattle computer vision literature, with nine papers that are directly addressing automated BCS assessment. Liang et al. (2025) **liang2025bcs** achieved state-of-the-art performance on the ScienceDB BCS dataset using EfficientNet-B0 with Squeeze-and-Excitation attention blocks, demonstrating 93.77% accuracy on 5-class BCS prediction and highlighting the effectiveness of attention mechanisms for focusing on relevant anatomical features while using publicly available data that makes this work particularly relevant as a baseline for comparison. Zheng et al. (2024) **zheng2024bcsyolo** proposed BCS-YOLO, combining YOLO-based detection with knowledge distillation for efficient BCS prediction, achieving 92.7% accuracy on the ScienceDB dataset while maintaining inference speeds suitable for real-time applications through model compression without significant accuracy loss. Li et al. (2025) **li2025yolov5bcs** developed an improved YOLOv5s architecture for BCS, achieving 91.8% accuracy on a private 5-class dataset, and while the private data limits reproducibility, the architectural modifications including enhanced feature pyramid networks provide valuable design insights. Summerfield et al. (2023) **summerfield2023depth** investigated depth-based BCS using multiple 3D cameras in a late fusion CNN architecture, achieving only 35.77% accuracy, which highlights the challenges of multi-camera depth fusion and the importance of careful sensor calibration while suggesting that naive depth fusion may not outperform well-designed RGB approaches. O’Mahony et al. (2023) **omahony2023rgbd** explored RGB-D fusion using MobileNet for RGB features and PointNet for depth point clouds, achieving 53% accuracy on 5-class BCS, demonstrating the complexity of multimodal fusion but validating the potential value of depth information when properly integrated. Rodenburg et al. (2024) **rodenburg2024ordinal** employed EfficientNetV2 with ordinal regression, explicitly modeling the ordered nature of BCS classes and achieving 84.6% accuracy using a private dataset, demonstrating the advantage of ordinal approaches over standard classification for BCS prediction. Gjergji et al. (2022) **gjergji2022weight** developed a multi-input CNN processing RGB-D data for combined weight and BCS estimation, achieving 91.9% BCS accuracy (as regression R^2) on private data and demonstrating the synergy between weight and body condition assessment. Lewis et al. (2025) **lewis2025yolov8bcs** applied YOLOv8 for BCS detection and classification, achieving 43.9% mAP on a 3-class problem, with the lower accuracy reflecting the challenging 3-class formulation and object detection framing rather than pure classification. Santos et al. (2025) **santos2025rf** employed traditional Random Forest classifiers on RGB image features, achieving $R^2=0.40$ for BCS regression, providing a baseline that demonstrates deep learning approaches significantly outperform traditional machine learning for BCS assessment.

2.2.2 Lameness Detection Research

Lameness detection research mostly utilizes video-based analysis with constitutes estimation as well as gait analysis pipelines. Tun et al. (2024) **tun2024depth** achieved remarkable 99.94% accuracy using top-view depth cameras with Detectron2

Table 2.1: Summary of Body Condition Scoring Papers

Reference	Architecture	Input	Accuracy	Dataset
Liang 2025	EfficientNet-B0+SE	RGB	93.77%	ScienceDB
Zheng 2024	BCS-YOLO + KD	RGB	92.7%	ScienceDB
Li 2025	Improved YOLOv5s	RGB	91.8%	Private
Gjergji 2022	Multi-input CNN	RGB-D	91.9%	Private
Rodenburg 2024	EfficientNetV2	RGB	84.6%	Private
O’Mahony 2023	MobileNet/PointNet	RGB+D	53%	Private
Lewis 2025	YOLOv8	RGB	mAP 43.9%	Private
Santos 2025	Random Forest	RGB	R ² =0.40	Private
Summerfield 2023	Late Fusion CNN	Depth	35.77%	Private

for detection and SORT for tracking, with the exceptional performance demonstrating the value of depth imaging for lameness detection when appropriate viewing angles are available. Myint et al. (2024) **myint2024lameness** developed a real-time lameness detection system using Mask R-CNN combined with machine learning classifiers on extracted gait features, achieving 98.98% accuracy from side-view RGB video and demonstrating that RGB-only solutions can achieve high performance with proper feature engineering. Kang et al. (2020) **kang2020lameness** proposed RFB-NET-SSD for cattle detection followed by gait phase analysis, achieving 96% lameness classification accuracy, with the supporting phase analysis identifying when each hoof contacts the ground and providing interpretable features for veterinary validation. Barney et al. (2023) **barney2023multipose** employed Mask R-CNN with SORT tracking for multi-cattle pose estimation, achieving 94% lameness detection accuracy, demonstrating that the ability to simultaneously track and assess multiple animals is crucial for practical farm deployment. Russello et al. (2024) **russello2024zed** combined RGB-D data from ZED cameras with pose estimation and traditional classifiers (Random Forest, SVM), achieving 80.07% accuracy, with the evaluation of multiple locomotion traits beyond binary lameness providing richer assessment capabilities.

Table 2.2: Summary of Lameness Detection Papers

Reference	Architecture	Input	Accuracy	Innovation
Tun 2024	Detectron2 + SORT	Depth	99.94%	Top-view depth
Myint 2024	Mask R-CNN + ML	RGB video	98.98%	Real-time side-view
Kang 2020	RFB-NET-SSD	RGB video	96%	Gait phase analysis
Barney 2023	Mask R-CNN + SORT	RGB video	94%	Multi-cattle pose
Russello 2024	Pose + RF/SVM	RGB-D	80.07%	Multiple traits

2.2.3 Behavior Recognition Research

Behavior recognition covers vision-based as well as sensor-based approaches, with MmCows becoming the only public dataset in this classification. Rohan et al. (2024) **rohan2024review** conducted a systematic review of 44 papers on cattle behavior recognition, identifying key trends including the dominance of YOLO-based detection, increasing use of deep learning, and the challenge of limited public datasets. Petcu et al. (2023) **petcu2023cnnlstm** achieved 97.35% accuracy using CNN-LSTM architecture on private RGB video data, with the combination of CNN for spatial features and LSTM for temporal modeling representing a standard approach for video-based behavior classification. Lee et al. (2023) **lee2023hanwoo** developed a pipeline combining YOLOv5 detection, SORT tracking, and I3D temporal classification, achieving 95.3% accuracy on private Hanwoo cattle data, with the I3D (Inflated 3D ConvNet) architecture effectively capturing spatiotemporal patterns in behavior. Giannone et al. (2025) **giannone2025behavior** employed YOLOv8 with region-of-interest extraction for behavior recognition, achieving 72% accuracy on private data, with the lower accuracy reflecting the complexity of fine-grained behavior discrimination. Dottorini et al. (2025) **dottorini2025imu** achieved 91.8% accuracy using XGBoost on Discrete Wavelet Transform features extracted from IMU sensor data in the MmCows dataset, demonstrating the value of sensor-based approaches and providing a public benchmark for comparison.

Table 2.3: Summary of Behavior Recognition Papers

Reference	Architecture	Input	Accuracy	Dataset
Petcu 2023	CNN-LSTM	RGB video	97.35%	Private
Lee 2023	YOLOv5+SORT+I3D	RGB video	95.3%	Private
Dottorini 2025	XGBoost + DWT	IMU sensor	91.8%	MmCows
Giannone 2025	YOLOv8 + ROI	RGB video	72%	Private
Rohan 2024	Systematic Review	Various	N/A	44 papers

2.2.4 Multi-Task Learning and Survey Papers

Numerous papers gives foundational knowledge for multi-task learning approaches also comprehensive surveys of agricultural AI. Crawshaw (2020) **crawshaw2020mtl** published a highly influential MTL survey, establishing a taxonomy of MTL approaches with key findings relevant to this research including that hard parameter sharing (shared backbone) is effective for heterogeneous tasks, sequential task training prevents negative transfer between conflicting tasks, task weighting strategies significantly impact multi-task performance, and frozen backbone approaches reduce computational requirements during training. Shi et al. (2023) **shi2023progressive** proposed progressive parameter sharing for MTL, dynamically adjusting sharing ratios during training, with this approach addressing the limitation of fixed sharing architectures that may not suit all task combinations. Attri et al. (2025) **attri2025saamvetnet** developed SAAM-VetNet, an attention-based multi-task CNN for simultaneous disease classification and severity assessment, demonstrating successful MTL in veterinary applications though focused on general animal disease

rather than cattle-specific monitoring. Carraro et al. (2026) **carraro2026cleanliness** investigated cattle cleanliness assessment using RGB and thermal fusion, critically noting that this paper explicitly states that no public cleanliness-annotated cattle datasets exist, requiring private data collection, which directly informs our decision to exclude cleanliness from the current research scope. Tzanidakis et al. (2024) **tzanidakis2024goatcnn** developed Goat-CNN for pose-independent BCS scoring in goats, and while targeting a different species, the lightweight architecture design and pose-invariance strategies are transferable to cattle applications.

Table 2.4: Summary of MTL, Survey, and Other Papers

Reference	Focus	Type	Key Contribution
Crawshaw 2020	Multi-Task Learning	Survey	MTL taxonomy and guidelines
Shi 2023	Progressive Sharing	Method	Dynamic parameter sharing
Attri 2025	Disease + Severity	Multi-task	Attention-based veterinary MTL
Carraro 2026	Cleanliness	Multimodal	RGB+Thermal (private data)
Tzanidakis 2024	Goat BCS	Lightweight	Pose-independent scoring

2.3 Summary of Key Findings

The structured review of 29 papers reveals many critical insights that directly inform our proposed methodology.

The substantial majority of papers (89%) rely on private datasets which creates a reproducibility crisis. Among them only a small fraction relies on publicly available cattle-specific datasets, yet they still obtain competitive performance, indicating that public datasets can efficiently support high-quality research. Our targeted use of public datasets helps bridge this gap, enabling total reproducibility as well as fair benchmarking.

Despite being consistently raised on multi-task learning in most of the literature, no existing paper integrates BCS, behavior recognition, lameness detection, and individual identification in a unified framework. Most of the papers address single tasks in isolation. Our unified four task framework represents a true novelty with potential for improved efficiency and cross task learning.

Transfer learning with frozen backbone is highly analysed in the literature, and the Crawshaw 2020 MTL Survey validates that this approach effectively prevents negative transfer between heterogeneous tasks. Our sequential task training strategy is well-supported by existing literature and provides a strong foundation for managing multiple diverse tasks.

Depth imaging represents substantial research interest with demonstrated value for BCS as well as lameness. Still, public data is limited to MmCows and Dryad datasets. Depth modality is relevant for BCS using Dryad but should be treated as secondary to RGB, ensuring the framework remains accessible for standard camera deployments.

Self-supervised learning remains highly underexplored in cattle monitoring compared to transfer learning, representing a potential research opportunity. MmCows provides 2 weeks of continuous unlabeled video ideal for SSL pre-training. SSL represents a potential bonus contribution if time permits, offering opportunities to leverage abundant unlabeled surveillance footage.

Comprehensive search reveals there are no publicly available thermal/infrared datasets for cattle. The only thermal cattle research (Carraro 2026) used privately collected data. Thermal modality is excluded from current scope and documented as future work requiring data collection partnerships.

Table 2.5: Summary of Key Cattle Monitoring Techniques from Literature

Technique	Platform	Accuracy	Strength	Weakness
EfficientNet+SE	RGB BCS	93.77%	High accuracy, attention	Single task
BCS-YOLO+KD	RGB BCS	92.7%	Edge-ready, distilled	Single task
Detectron2+SORT	Depth Lameness	99.94%	Excellent detection	Requires depth
CNN-LSTM	RGB Behavior	97.35%	Temporal modeling	Private data
XGBoost+DWT	IMU Behavior	91.8%	Public benchmark	Requires sensors
MTL Survey	Cross-platform	N/A	Guidelines	No implementation

Chapter 3

Conclusion

The research which was presented in this pre-thesis 1 report aims to develop a multi-task deep learning framework for unified cattle health and behavior monitoring using public RGB and depth datasets. We are hopeful that this research will contribute significantly to precision livestock farming. Working with diverse cattle health monitoring tasks is highly challenging due to the complexity of integrating multiple objectives and the limited processing capacity of edge devices. Our framework strives to robustly monitor and protect cattle health indicators from potential adversarial scenarios and corrupted data. Our research does not cover everything; there is huge ground to cover. Agricultural sectors are rapidly evolving and slowly blurring the line between traditional and smart farming. Collaborative and federated learning approaches can serve as great solutions for cattle health monitoring. So, protecting these models from attacks and data corruption should be considered a top priority. The integration of RGB and depth modalities, combined with MTL, provides us with a strong approach in the way of automating cattle health assessments that improve animal welfare and productivity. In the future, we will focus on the deployment of these models in real-world environments to validate performance across different cattle breeds and conditions, and explore federated learning for secure collaborative training without centralizing data while preserving data privacy.